Data-Driven Forecasting & Peak Demand Prediction

Afficionado Coffee Roasters

Exploratory Data Analysis (EDA), demand-pattern assessment, forecasting-model review, and business insights

Transactions	Quantity Sold	Revenue	Avg. Transaction Value
149,116	**214,470**	**$698,812.33**	**$4.69**

Source basis: the supplied Jupyter notebook and supplied Power BI (.pbix) project. The notebook contains 149,116 transaction rows and the PBIX contains five report pages: Home Page, Executive Overview, Peak Demand & Operations, Product & Store Analysis, and Forecasting & Demand Planning.

Important scope note: the source data contains year and time-of-day but no true transaction date. The notebook therefore creates a transaction date by combining the year with the time, which collapses records to 2025-01-01. This supports intraday EDA, but it does not support genuine day-by-day or seasonal forecasting from the supplied data.

1. Executive Summary

The dataset contains **149,116 transactions**, **214,470 units**, and **$698,812.33** in recorded revenue. The average transaction value is **$4.69**. The analysis indicates that demand is strongly concentrated around the morning operating window, making intraday staffing, replenishment and production planning important operational levers.

At store level, revenue is relatively balanced across the three locations. Hell's Kitchen records $236,511.17 (33.8%), Astoria records $232,243.91 (33.2%), and Lower Manhattan records $230,057.25 (32.9%). Lower Manhattan has the highest average transaction value among the three stores in the notebook output.

Product mix is more concentrated: Coffee contributes 38.6% of revenue and Tea contributes 28.1%. Together they account for 66.7% of recorded revenue. This suggests that availability of core beverage categories is especially important during high-demand periods.

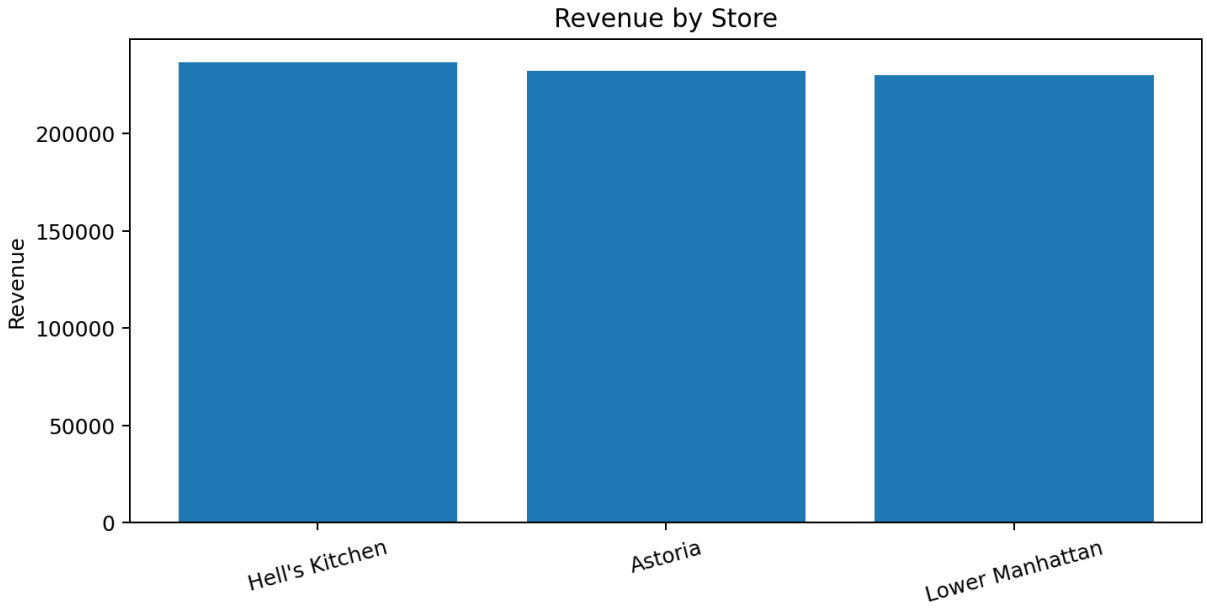
Forecasting interpretation: the notebook reports a machine-learning demand model with MAE = 0.4365 and RMSE = 0.4899 on its held-out records. However, the train/test split is based on a row ordering sorted by hour and minute rather than a chronological calendar split. Therefore these metrics should be treated as model-development evidence for the supplied analysis, not as a validated future-demand forecast accuracy measure.

2. Data Profile & Preparation

Item	Source finding
Rows	149,116 transaction records
Columns	11 source fields
Year	2025
Core measures	transaction quantity, unit price, revenue
Operational dimensions	store, store location, product, product category, transaction time
Derived fields	transaction datetime, hour, minute, daypart, revenue
Duplicate handling	Duplicate transaction_id values removed
Missing-value handling	Rows with missing transaction_id, datetime, quantity or unit price removed

The notebook's quality check reports zero missing values in the cleaned analytical fields. This is a strong data-quality result for the supplied file. The main analytical limitation is not missingness but the absence of a true calendar date.

3. Store-Level EDA

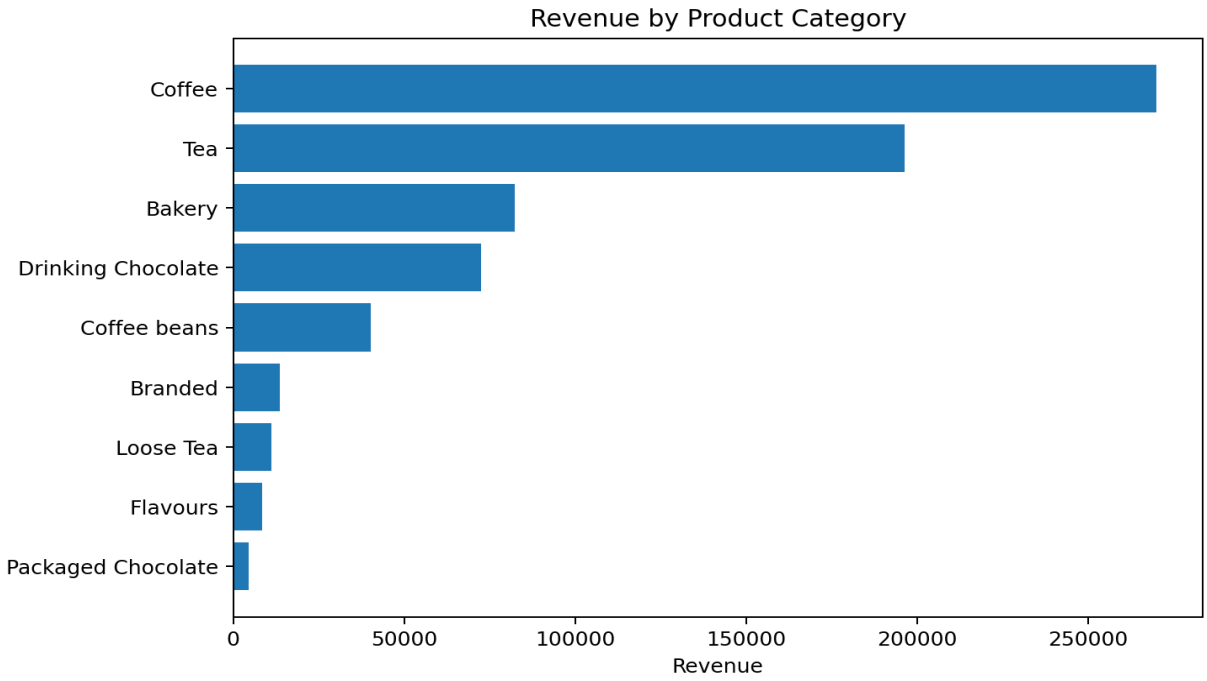


Store	Transactions	Quantity	Revenue	ATV	Revenue Share
Hell's Kitchen	50,735	71,737	$236,511.17	$4.66	33.8%
Astoria	50,599	70,991	$232,243.91	$4.59	33.2%
Lower Manhattan	47,782	71,742	$230,057.25	$4.81	32.9%

Business insight: store revenue is highly balanced, with each location contributing roughly one-third of total revenue. This reduces dependence on a single store and suggests that demand-planning standards can be shared across locations while still allowing store-specific peak-hour adjustments.

Operational implication: Lower Manhattan has the highest average transaction value ($4.81 in the notebook output), while Hell's Kitchen generates the highest total revenue. Management can therefore distinguish between traffic-driven revenue and basket-value-driven revenue when evaluating store performance.

4. Product & Category EDA

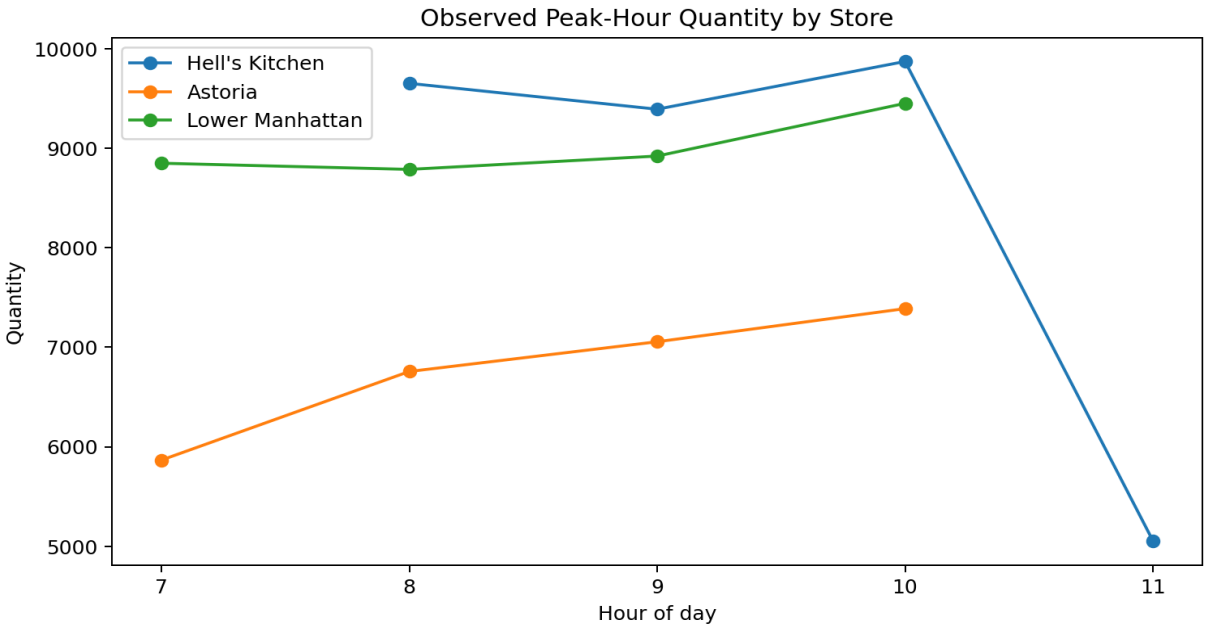


Category	Transactions	Quantity	Revenue	Revenue Share
Coffee	58,416	89,250	$269,952.45	38.6%
Tea	45,449	69,737	$196,405.95	28.1%
Bakery	22,796	23,214	$82,315.64	11.8%
Drinking Chocolate	11,468	17,457	$72,416.00	10.4%
Coffee beans	1,753	1,828	$40,085.25	5.7%
Branded	747	776	$13,607.00	1.9%
Loose Tea	1,210	1,210	$11,213.60	1.6%
Flavours	6,790	10,511	$8,408.80	1.2%
Packaged Chocolate	487	487	$4,407.64	0.6%

Business insight: Coffee and Tea together generate 66.7% of total revenue. These categories should receive priority in demand planning, replenishment and peak-hour availability monitoring.

Portfolio insight: Bakery and Drinking Chocolate are meaningful secondary categories, while Coffee beans, Branded products, Loose Tea, Flavours and Packaged Chocolate form smaller revenue pools. The smaller categories may still be useful for cross-selling, but the supplied EDA does not establish causal relationships or profitability.

5. Peak Demand & Intraday Pattern Analysis



The notebook defines store-level peak hours using the 75th percentile of hourly quantity within each store. The resulting peak-hour table shows a clear concentration in the morning period.

Store	Peak hour	Quantity	Transactions	Revenue
Astoria	10:00	7,388	5,291	$24,426.12
Astoria	09:00	7,055	5,083	$23,183.57
Astoria	08:00	6,756	4,966	$22,805.90
Astoria	07:00	5,865	4,181	$19,028.80
Hell's Kitchen	10:00	9,873	6,957	$33,605.81
Hell's Kitchen	08:00	9,653	6,909	$31,544.44
Hell's Kitchen	09:00	9,393	6,767	$32,874.29
Hell's Kitchen	11:00	5,053	3,598	$17,926.54
Lower Manhattan	10:00	9,452	6,297	$30,641.46
Lower Manhattan	09:00	8,922	5,914	$29,111.67
Lower Manhattan	07:00	8,850	5,792	$28,536.62
Lower Manhattan	08:00	8,788	5,779	$28,349.53

Peak-demand insight: Hell's Kitchen reaches 9,873 units at 10:00 in the peak table, Lower Manhattan reaches 9,452 units at 10:00, and Astoria reaches 7,388 units at 10:00. The repeated 07:00–10:00 concentration indicates that the morning rush should be treated as a key operating window.

Recommended operational response: prepare core beverage inventory before the morning rush, schedule stronger front-of-house coverage in the identified peak window, and monitor stockouts or service-time increases specifically around these hours.

6. Forecasting & Demand-Prediction Assessment

Model component	Notebook implementation
Target	transaction_qty
Features	unit_price, hour, minute, store_location, product_category, product_type
Categorical encoding	OneHotEncoder(handle_unknown='ignore')
Model	HistGradientBoostingRegressor
Iterations	150
Split	80% train / 20% test
Reported MAE	0.4365
Reported RMSE	0.4899

The notebook's model is useful as a starting point for intraday quantity prediction because it combines time-of-day, store and product information. The reported MAE of 0.4365 means the average absolute prediction error on the notebook's held-out records is about 0.44 transaction units under that evaluation setup.

Validation caution: the notebook creates transaction_datetime from year + transaction_time and then sorts the model data by hour and minute. It does not perform a true chronological train/test split. In addition, the source data does not contain a genuine transaction date. Consequently, the current model should not be presented as a validated multi-day or seasonal forecast model.

Next modelling step: obtain a real transaction date, aggregate demand by date × store × product/category or hour × store, then create lag features, rolling averages, day-of-week, holiday and promotion variables. A chronological backtest should then be used to evaluate future-demand accuracy.

7. Business Insights & Actions

1. Protect the morning revenue window

Peak quantities repeatedly occur between 07:00 and 10:00 across stores. Pre-position core inventory and align staffing with this window.

2. Prioritize Coffee and Tea availability

Coffee and Tea together contribute 66.7% of revenue in the supplied category summary. These categories deserve the strongest replenishment and stockout monitoring.

3. Use store-specific planning

Store revenue is balanced, but hourly peak volumes differ. A single staffing or replenishment rule may therefore be less precise than store-specific thresholds.

4. Monitor basket value separately from traffic

Lower Manhattan has the highest average transaction value, while Hell's Kitchen has the highest revenue. Store management can track transaction count and basket value as separate performance drivers.

5. Upgrade forecasting from intraday prediction to true time-series forecasting

The present model predicts transaction quantity using time and categorical features, but the data lacks a real date. A dated history is required to learn weekday, seasonal, holiday and promotional demand patterns.

6. Turn the 75th-percentile rule into an operational alert

The notebook already defines peak hours using a store-level 75th percentile. This can be converted into Power BI alerts or daily operating thresholds once dated data is available.

Suggested KPI set for management

KPI	Purpose
Hourly Quantity	Monitor intraday demand pressure
Peak-Hour Quantity	Size operational load in the busiest window
Peak-Hour Revenue	Estimate revenue exposure during demand peaks
Average Transaction Value	Monitor basket-value performance
Category Revenue Share	Track dependence on core categories
Forecast MAE / RMSE	Monitor prediction accuracy after proper time-based validation
Stockout Rate during Peak Hours	Connect demand prediction to operational execution

8. Data & Analytical Limitations

• **No true transaction date:** the source contains year and transaction time but not a calendar date. The notebook-generated transaction_date therefore becomes 2025-01-01 for all rows.

• **No seasonality analysis:** without actual dates, weekday, month, holiday and seasonal patterns cannot be measured from the supplied transaction file.

• **Forecast validation is not chronological:** the model's 80/20 split follows an hour/minute ordering rather than a future-date holdout.

• **No external drivers:** weather, promotions, holidays, events and local footfall are not included in the supplied notebook analysis.

• **Profitability is not measured:** the report uses revenue, quantity and transaction value; margins, product costs and labor costs are not available in the supplied analysis.

• **Forecast target granularity:** transaction-level quantity prediction is different from operational daily/hourly demand forecasting. Aggregated demand forecasting should be tested separately.

9. Conclusion

The supplied analysis establishes a useful baseline for Afficionado Coffee Roasters: the business records substantial transaction volume, revenue is distributed fairly evenly across the three stores, Coffee and Tea dominate the revenue mix, and demand is concentrated in the morning operating window. These findings can directly support staffing, replenishment and peak-period readiness.

For the project to fully support the title *Data-Driven Forecasting & Peak Demand Prediction*, the most important data enhancement is a genuine transaction date. With that field, the current EDA can be extended into daily/hourly time-series forecasting with chronological backtesting, lag/rolling features and calendar effects. Until then, the current results should be framed as intraday demand analysis and a preliminary prediction-model baseline.

Appendix: Power BI Project Structure

The supplied PBIX report contains five report pages, identified from its internal report layout:

• Home Page

• Executive Overview page

• Peak Demand & Operations page

• Product & Store Analysis page

• Forecasting & Demand Planning page

Source files reviewed: supplied Jupyter notebook (.ipynb) and supplied Power BI report (.pbix). All numerical findings in this report are derived from the notebook outputs available in the supplied files; no external dataset was substituted.